# Edexcel International AS Chemistry

Your notes

## Group 7

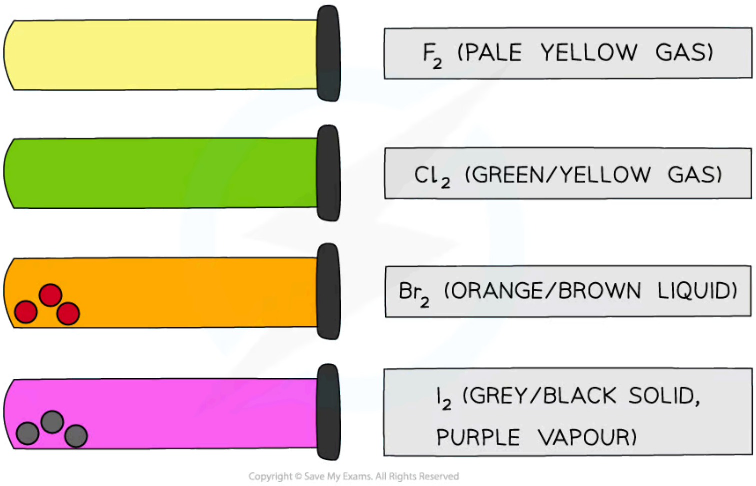
### Contents

# Trends of the Group 7 Elements

- The Group 7 elements are called **halogens**
- The halogens have uses in water purification and as bleaching agents (chlorine), as flame-retardants and fire extinguishers (bromine) and as antiseptic and disinfectant agents (iodine)

## Colours

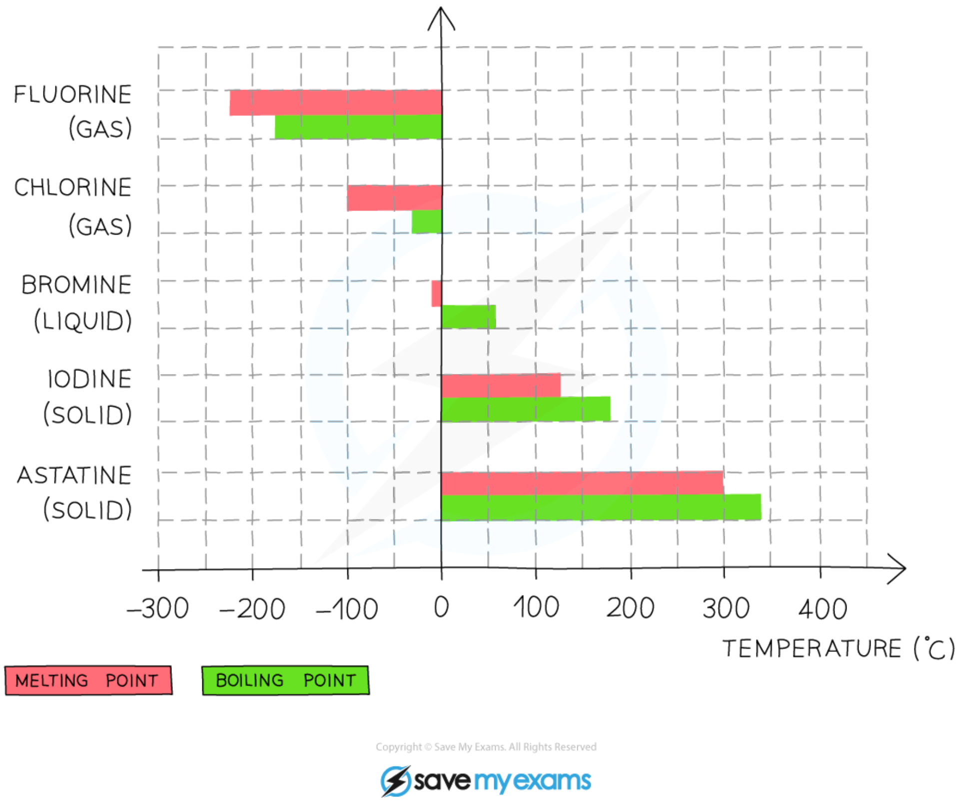
- All halogens have distinct **colours** which get **darker** going down the group




*The colours of the Group 7 elements get darker going down the group*

## Volatility

- **Volatility** refers to how easily a substance can evaporate
  - A volatile substance will have a low boiling point




***The melting and boiling points of the Group 7 elements increase going down the group which indicates that the elements become less volatile***

- Going down the group, the **boiling point** of the elements increases which means that the **volatility** of the halogens decreases
    - This means that fluorine is the most volatile and iodine the least volatile

## Trend in melting and boiling points

- Halogens are non-metals and are **diatomic molecules** at room temperature
    - This means that they exist as molecules which are made up of two similar atoms, such as $F_2$
- The halogens are **simple molecular structures** with **weak** London dispersion forces between the diatomic molecules caused by instantaneous dipole-induced dipole forces

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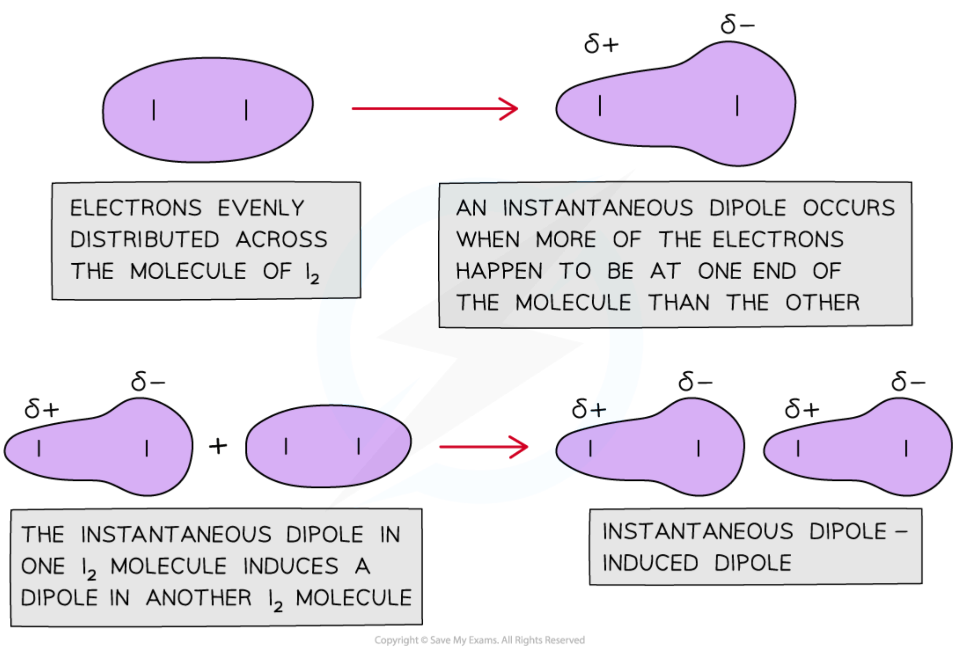
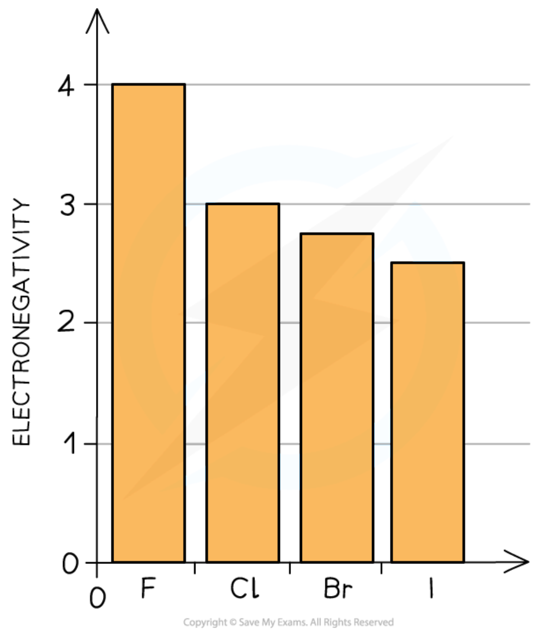



***The diagram shows that a sudden imbalance of electrons in a nonpolar molecule can cause an instantaneous dipole. When this molecule gets close to another non-polar molecule it can induce a dipole as the cloud of electrons repel the electrons in the neighbouring molecule to the other side***

- The more **electrons** there are in a molecule, the greater the instantaneous dipole-induced dipole forces
- Therefore, the **larger** the molecule the **stronger** the London dispersion forces between molecules
- This is why as you go down the group, it gets more difficult to separate the molecules and the **melting** and **boiling points** increase
- As it gets more difficult to separate the molecules, the **volatility** of the halogens **decreases** going down the group

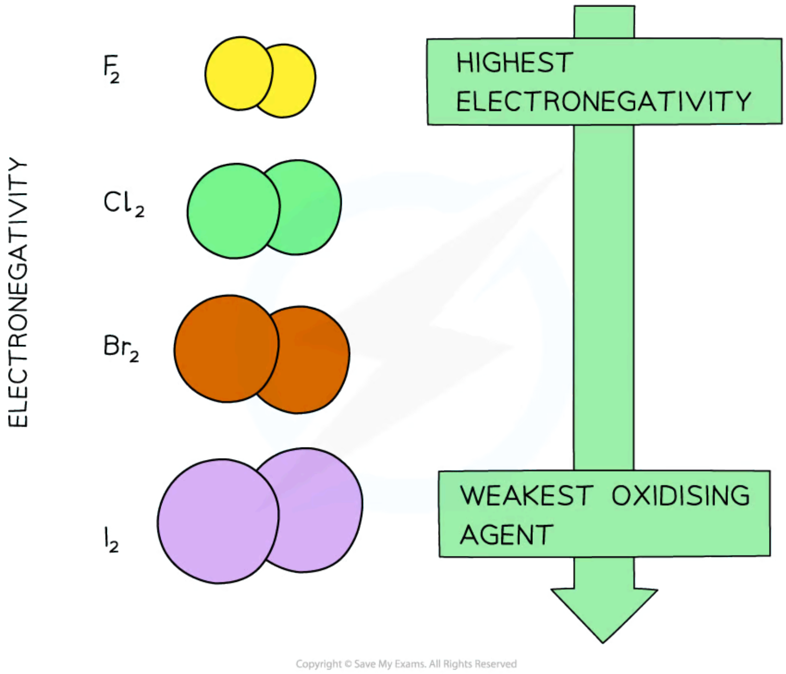
## Trend in electronegativity

- The electronegativity of of the halogens decreases down the group




***The electronegativity of the halogens decreases going down the group***

- The **electronegativity** of an atom refers to how strongly it attracts electrons towards itself in a covalent bond
- The decrease in electronegativity is linked to the size of the halogens
- Going down the group, the atomic radii of the elements increase which means that the outer shells get further away from the nucleus
- An 'incoming' electron will therefore experience more **shielding** from the attraction of the positive nuclear charge
- The halogens' ability to accept an electron (their **oxidising power**) therefore decreases going down the group




***With increasing atomic size of the halogens (going down the group) their electronegativity, and therefore oxidising power, decreases***

## Reactivity

- When a halogen atom reacts it will usually gain an electron, to form a **1–** ion ($X + e^- \rightarrow X^-$)
- The oxidation number has decreased from 0 to -1, therefore reduction has occurred
- Therefore halogens will act as oxidising agents
- Down Group 7 we have seen that the atoms become larger so the outer electrons are further away and are therefore more shielded from the positive nucleus
- Larger halogen atoms such as iodine will find it more difficult to attract incoming electrons needed to form the 1– ion
- Therefore the reactivity decreases down Group 7

## Reaction with hydrogen

- To demonstrate the decrease in the reactivity we can look a the reaction with hydrogen gas
- The table outlines the trend in the reactivity of the halogens with hydrogen gas
- As we can see the reaction becomes less vigorous down the group

## Reaction between Halogen & Hydrogen Gas

| Equation for Reaction | Description of Reaction |
|---|---|
| $H_2(g) + F_2(g) \longrightarrow 2HF(g)$ | Reacts explosively even in cool, dark conditions |
| $H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$ | Reacts explosively in sunlight |
| $H_2(g) + Br_2(g) \longrightarrow 2HBr(g)$ | Reacts slowly on heating |
| $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ | Forms an equilibrium mixture on heating |



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## Halogen Displacement Reactions

# Halogen Displacement

- The reactivity of halogens is also shown by their **displacement reactions** with other halide ions in solutions
  - A **more reactive** halogen can displace a **less reactive** halogen from a halide solution of the less reactive halogen
  - Eg. The addition of chlorine water to a solution of bromine water:

**$Cl_2$ (aq) + 2NaBr (aq) → 2NaCl (aq) + $Br_2$ (aq)**

  - The chlorine has displaced the bromine from solution as it is more reactive which can be summarised in the following ionic equation by removing the sodium spectator ions:

$$Cl_2\,(aq) + 2Br^-\,(aq) \rightarrow 2Cl^-\,(aq) + Br_2\,(aq)$$

  - Chlorine can also displace iodine from a solution of iodide ions:

$$Cl_2\,(aq) + 2I^-\,(aq) \rightarrow 2Cl^-\,(aq) + I_2\,(aq)$$

- We can see these are both redox reactions by taking a look at changes in the oxidation number of each element in the reaction
  - Br and I both change from = -1 → 0 so the bromine and iodine have been oxidised in their respective reactions
  - Cl = 0 → -1 so the chlorine has been reduced in both reactions
  - No change in oxidation number for the sodium

## Chlorine with Bromides & Iodides

- If you add chlorine solution to colourless potassium bromide or potassium iodide solution a displacement reaction occurs:
  - The solution becomes **yellow-orange** as bromine is formed
  - The solution becomes **brown** as iodine is formed
- If an **organic solvent** is added, such as cyclohexane, the following observations are seen:
  - The organic layer will appear **yellow-orange** as bromine is formed
  - The organic layer will appear **purple** as iodine is formed
  - The organic solvent is useful as the the halogens are more soluble in this layer which helps observe the colour changes more easily
- Chlorine is **above** bromine and iodine in Group 7 so it is more reactive
- Chlorine will **displace** bromine or iodine from an aqueous solution of the metal halide:

$$Cl_2\,(aq) + 2KBr\,(aq) \rightarrow 2KCl\,(aq) + Br_2\,(aq)$$

chlorine + potassium bromide → potassium chloride + bromine

$$Cl_2\,(aq) + 2KI\,(aq) \rightarrow 2KCl\,(aq) + I_2\,(aq)$$

chlorine + potassium iodide → potassium chloride + iodine

## Bromine with Iodides

- Bromine is above iodine in Group 7 so it is **more** reactive
- Bromine will displace iodine from an aqueous solution of the metal iodide

bromine + potassium iodide → potassium bromide + iodine

$$Br_2\,(aq) + 2KI\,(aq) \rightarrow 2KBr\,(aq) + I_2\,(aq)$$

- The reaction will turn **brown** as iodine is formed
- If an organic solvent is added the organic layer will appear **purple** as iodine is formed
- We can show that this is a redox reaction by looking at the changes in oxidation numbers:
  - I = -1 → 0 so the iodine has been oxidised
  - Br = 0 → -1 so the bromine has been reduced
  - No change in oxidation number for the potassium
- Rather than writing the full equation, we can also write the ionic equation by removing the potassium spectator ion

$$Br_2\,(aq) + 2I^-\,(aq) \rightarrow 2Br^-\,(aq) + I_2\,(aq)$$

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# Redox Reactions of the Halogens

## Reactions with Group 1 & 2 metals

- The halogens react with some metals to form **ionic compounds** which are **metal halide salts**
- In all reactions where halogens are reacting with metals, the metals are being oxidised
- Reaction of sodium and chlorine
  - $2Na\,(s) + Cl_2\,(g) \rightarrow 2NaCl\,(s)$
  - Na is being oxidised, the oxidation number is changing from 0 to +1
- Calcium is a group 2 metal:
  - $Ca\,(s) + Br_2\,(l) \rightarrow CaBr_2\,(s)$
  - Ca is being oxidised, the oxidation number is changing from 0 to +2
- Therefore the halogens are acting as oxidising agents

## Reactions with Iron(II)

- Chlorine and bromine can oxidise iron(II) to iron(III)

$$Cl_2\,(g) + 2Fe^{2+}\,(aq) \rightarrow 2Cl^-\,(aq) + 2Fe^{3+}\,(aq)$$

$$Br_2\,(g) + 2Fe^{2+}\,(aq) \rightarrow 2Br^-\,(aq) + 2Fe^{3+}\,(aq)$$

- However, iodine is not a strong enough oxidising agent to oxidise iron(II) to iron(III)
- Iodine is actually oxidised from iodide ions to iodine by iron(III)

$$2I^-\,(aq) + 2Fe^{3+}\,(aq) \rightarrow I_2\,(aq) + 2Fe^{2+}\,(aq)$$

## Disproportionation reaction

- A **disproportionation reaction** is a reaction in which the same species is both oxidised and reduced
- The reaction of **chlorine** with **dilute alkali** is an example of a disproportionation reaction
- In these reactions, the chlorine gets oxidised and reduced at the same time
- Different reactions take place at different **temperatures** of the dilute alkali

## Chlorine in cold alkali (15 °C)

- The reaction that takes place is:

$$Cl_2(aq) + 2NaOH\,(aq) \longrightarrow NaCl\,(aq) + NaClO\,(aq) + H_2O(l)$$

- The ionic equation is:

$$\underset{0}{Cl_2(aq)} + 2OH^-(aq) \longrightarrow \underset{-1}{Cl^-(aq)} + \underset{+1}{ClO^-(aq)} + H_2O(aq)$$

REDUCTION

OXIDATION



- The ionic equation shows that the chlorine gets both oxidised and reduced
- Chlorine gets oxidised as there is an increase in ox. no. from 0 to +1 in $ClO^-(aq)$
  - The half-equation for the oxidation reaction is:

$$\underset{0}{\tfrac{1}{2}Cl_2} + 2OH^-(aq) \longrightarrow \underset{+1}{ClO^-(aq)} + H_2O + e^-$$



- Chlorine gets reduced as there is a decrease in ox. no. from 0 to -1 in $Cl^-(aq)$
  - The half-equation for the reduction reaction is:

$$\underset{0}{\tfrac{1}{2}Cl_2} + e^- \longrightarrow \underset{-1}{Cl^-}$$



## Chlorine in hot alkali (70 °C)

- The reaction that takes place is:

$$3Cl_2(aq) + 6NaOH(aq) \longrightarrow 5NaCl(aq) + NaClO_3(aq) + 3H_2O(l)$$



- The ionic equation is:

$$\underset{0}{Cl_2(aq)} + 6OH^-(aq) \longrightarrow \underset{-1}{5Cl^-(aq)} + \underset{+5}{ClO_3^-(aq)} + H_2O(l)$$

REDUCTION

OXIDATION



- The ionic equation shows that the chlorine gets both oxidised and reduced

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- Chlorine gets oxidised as there is an increase in ox. no. from 0 to +5 in $ClO_3^-$(aq)
  - The half-equation for the oxidation reaction is:

$$\underset{0}{\tfrac{1}{2}Cl_2} + 6OH^- \longrightarrow \underset{+5}{ClO_3^-} + 3H_2O + e^-$$



- Chlorine gets reduced as there is a decrease in ox. no. from 0 to -1 in $Cl^-$(aq)
  - The half-equation for the reduction reaction is:

$$\underset{0}{2\tfrac{1}{2}Cl_2} + e^- \longrightarrow \underset{-1}{5Cl^-}$$



## Drinking water

- Chlorine can be used to clean water and make it drinkable
- The reaction of chlorine in water is a **disproportionation reaction** in which the chlorine gets both **oxidised** and **reduced**

$$\underset{0}{Cl_2(aq)} + H_2O(l) \longrightarrow \underset{-1}{HCl(aq)} + \underset{+1}{HClO(aq)}$$

REDUCTION

OXIDATION



***The disproportionation reaction of chlorine with water in which chlorine gets reduced to HCl and oxidised to HClO***

- Chloric(I) acid (HClO) **sterilises** water by killing **bacteria**
- Chloric acid can further dissociate in water to form $ClO^-$(aq):

$$\mathbf{HClO\ (aq) \rightarrow H^+\ (aq) + ClO^-\ (aq)}$$

- $ClO^-$(aq) also acts as a **sterilising agent** cleaning the water

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# Other Reactions of the Halogens

## Concentrated sulfuric acid

- Chloride, bromide and iodide ions react with concentrated sulfuric acid to produce **toxic gases**
- These reactions should therefore be carried out in a fume cupboard
- The general reaction of the halide ions with concentrated sulfuric acid is:

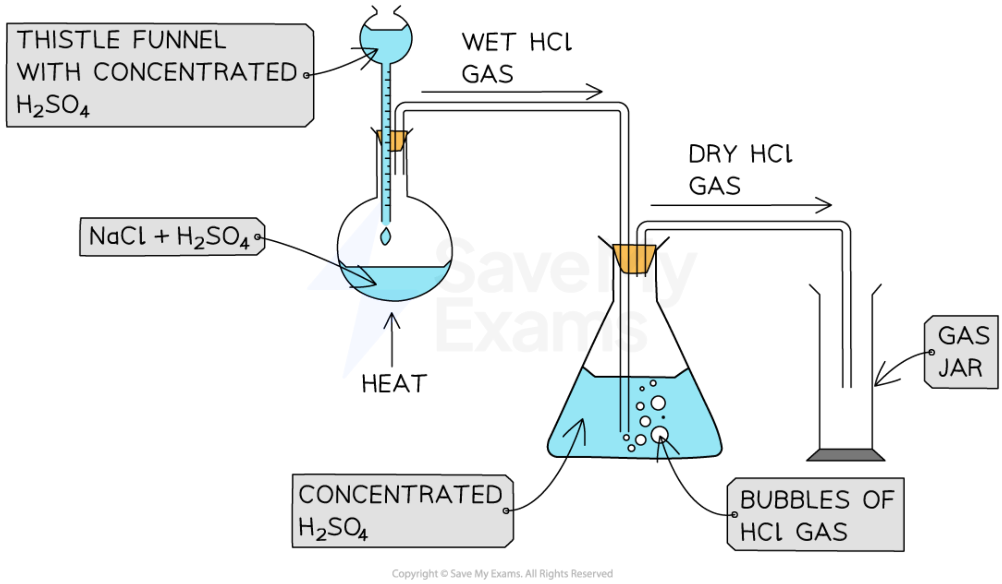
$$H_2SO_4(l) + X^-(aq) \rightarrow HX(g) + HSO_4^-(aq)$$

(general equation)

Where $X^-$ is the halide ion

## Reaction of chloride ions with concentrated sulfuric acid

- Concentrated sulfuric acid is dropwise added to sodium chloride crystals to produce **hydrogen chloride gas**




***Apparatus set up for the reaction of sodium chloride with concentrated sulfuric acid***

- The reaction that takes place is:

$$H_2SO_4\,(l) + NaCl\,(s) \rightarrow HCl\,(g) + NaHSO_4\,(s)$$

- The HCl gas produces is seen as **misty fumes**

## Reaction of bromide ions with concentrated sulfuric acid

- The **thermal stability** of the hydrogen halides decreases down the group

- The reaction of sodium bromide and concentrated sulfuric acid is:

$$H_2SO_4 (l) + NaBr (s) \rightarrow HBr (g) + NaHSO_4 (s)$$

- The concentrated sulfuric acid **oxidises** HBr which decomposes into **bromine** and **hydrogen gas** and sulfuric acid itself is **reduced** to **sulfur dioxide gas**:

$$2HBr (g) + H_2SO_4 (l) \rightarrow Br_2 (g) + SO_2 (g) + 2H_2O (l)$$

- The bromine is seen as a **reddish-brown gas**

## Reaction of iodide ions with concentrated sulfuric acid

- The reaction of sodium iodide and concentrated sulfuric acid is:

$$H_2SO_4 (l) + NaI (s) \rightarrow HI (g) + NaHSO_4 (s)$$

- **Hydrogen iodide** decomposes the **easiest**
- Sulfuric acid oxidises the hydrogen iodide to several extents:
- The concentrated sulfuric acid **oxidises** HI and is itself **reduced** to **sulfur dioxide gas**:

$$2HI (g) + H_2SO_4 (l) \rightarrow I_2 (g) + SO_2 (g) + 2H_2O (l)$$

- Iodine is seen as a violet/purple vapour
- The concentrated sulfuric acid **oxidises** HI and is itself **reduced** to **sulfur**:

$$6HI (g) + H_2SO_4 (l) \rightarrow 3I_2 (g) + S (s) + 4H_2O (l)$$

- Sulfur is seen as a **yellow solid**
- The concentrated sulfuric acid **oxidises** HI and is itself **reduced** to **hydrogen sulfide**:

$$8HI (g) + H_2SO_4 (l) \rightarrow 4I_2 (g) + H_2S (s) + 4H_2O (l)$$

- Hydrogen sulfide has a **strong smell of bad eggs**

## Further redox reaction involving sulfur dioxide

- HI is such a strong reducing agent that it can reduce sulfur dioxide, $SO_2$, further to hydrogen sulfide, $H_2S$

$$SO_2 (g) + 2HI (g) \rightarrow H_2S (g) + I_2 (s) + H_2O (l)$$

- Sulfur is reduced from +4 in $SO_2$ to –2 in $H_2S$, a gain of 6 electrons:
  - This demonstrates reduction
- The iodide ion is oxidised to iodine ($I_2$), seen as purple vapour
- The bad egg smell of $H_2S$ confirms the presence of this gas
- This final step highlights the maximum reducing power of iodide ions, not seen with $Cl^-$ or $Br^-$

## Halide ion Reactions with Concentrated Sulfuric Acid Table

Your notes

| Halide ion | Reaction with conc $H_2SO_4$ | Observations |
| --- | --- | --- |
| $Cl^-$ (aq) | $H_2SO_4$ (l) + NaCl (s) → HCl (g) + $NaHSO_4$ (s) | Misty fumes of HCl gas |
| $Br^-$ (aq) | $H_2SO_4$ (l) + NaBr (s) → HBr (g) + $NaHSO_4$ (s)<br><br>$H_2SO_4$ (l) + 2HBr (g) → $Br_2$ (g) + $SO_2$ (g) + $2H_2O$ (l) | Misty fumes of HBr gas<br><br>Chocking gas of $SO_2$ and reddish brown gas of $Br_2$ |
| $I^-$ (aq) | $H_2SO_4$ (l) + NaI (s) → HI (g) + $NaHSO_4$ (s)<br><br>$H_2SO_4$ (l) + 2HI (g) → $I_2$ (s) + $SO_2$ (g) + $2H_2O$ (l)<br><br>$H_2SO_4$ (l) + 6HI (g) → $3I_2$ (s) + S (s) + $4H_2O$ (l)<br><br>$H_2SO_4$ (l) + 8HI (g) → $4I_2$ (s) + $H_2S$ (g) + $4H_2O$ (l) | Misty fumes of HI gas<br><br>Chocking gas of $SO_2$ and purple vapour of $I_2$<br><br>Yellow solid of S (s)<br><br>Strong, bad egg smell of $H_2S$ (g) |

## Explaining the Trend in Reducing Ability of Halide Ions

- As you go down Group 7, the halide ions get bigger ($Cl^-$ → $Br^-$ → $I^-$)
- So, the attraction between the outer electrons and the nucleus gets weaker
- This makes it easier for the larger halide ions to lose electrons
- When they lose electrons, they reduce other substances,
- So the reducing power increases down the group:
  - $Cl^- < Br^- < I^-$

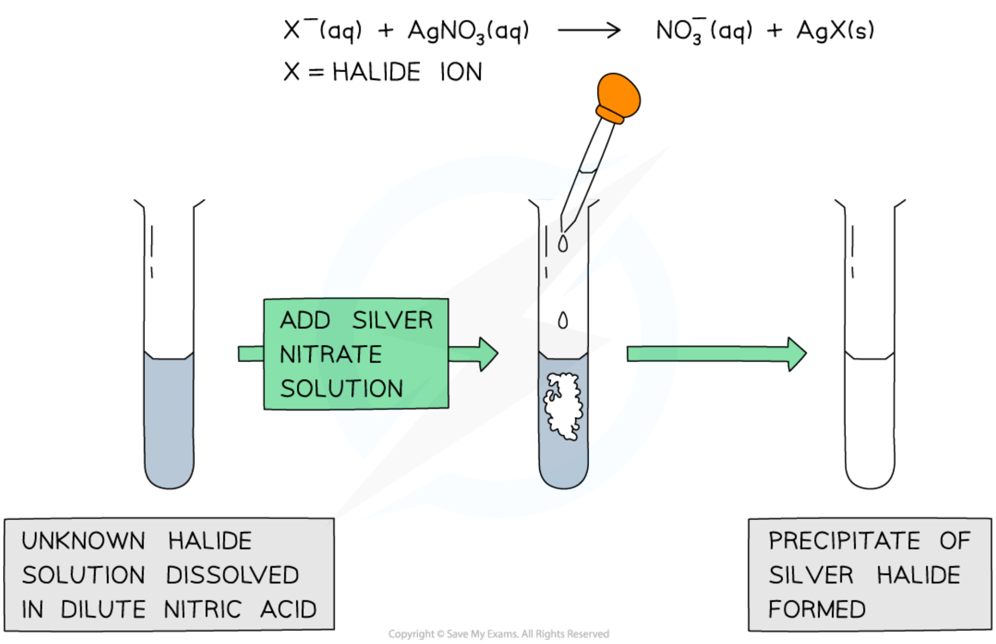
## Silver ions & ammonia

- Halide ions can be identified in an **unknown solution** by dissolving the solution in **nitric acid** and then adding a **silver nitrate solution** followed by **ammonia** solution
- The halide ions will react with the silver nitrate solution as follows:

$$\mathbf{AgNO_3\,(aq) + X^-\,(aq) \rightarrow AgX\,(s) + NO_3^-\,(aq)}$$

$$\mathbf{Ag^+\,(aq) + X^-\,(aq) \rightarrow AgX\,(s)}$$

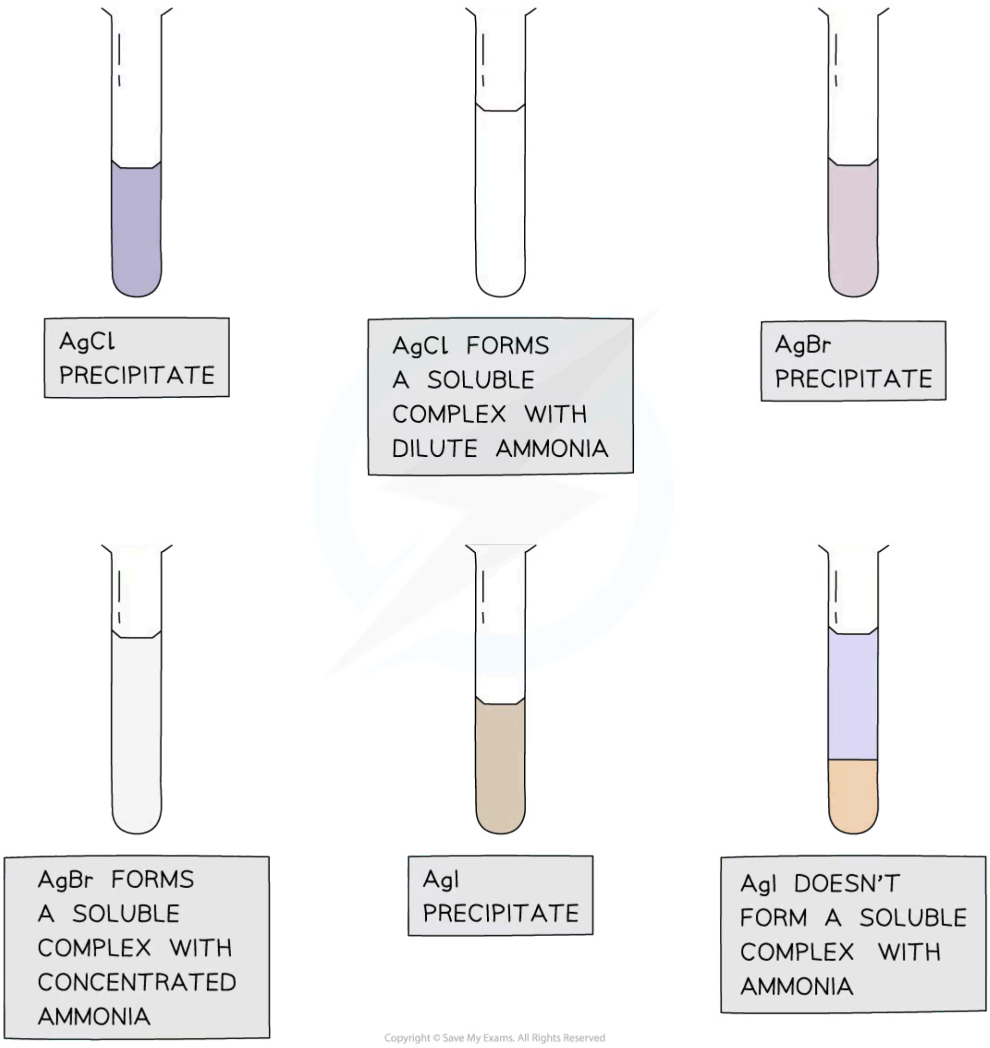
- $\mathbf{X^-}$ is the halide ion in both equations
- If the unknown solution contains halide ions, then a **precipitate** of the **silver halide** will be formed (AgX)




***A silver halide precipitate is formed upon addition of silver nitrate solution to halide ion solution***

- **Dilute** followed by **concentrated ammonia** is added to the silver halide solution to identify the halide ion
- If the precipitate dissolves in **dilute** ammonia the unknown halide is **chloride**
- If the precipitate does not dissolve in dilute but in **concentrated** ammonia the unknown halide is **bromide**
- If the precipitate does not dissolve in **dilute** nor **concentrated** ammonia the unknown halide is iodide

Your notes




***Silver chloride and silver bromide precipitates dissolve on addition of ammonia solution whereas silver iodide doesn't***

## Reaction of Halide Ions with Silver Nitrate & Ammonia Solutions Table

| Halide ion | Colour of Silver Halide Solution | Effect of adding dil. $NH_3$ | Effect of adding conc. $NH_3$ |
|---|---|---|---|
| $Cl^-$ (aq) | White | Dissolves | Dissolves |
| $Br^-$ (aq) | Cream | Insoluble | Dissolves |
| $I^-$ (aq) | Pale yellow | Insoluble | Insoluble |

## Reactions with hydrogen halides

Your notes

- When a halogen reacts with hydrogen a hydrogen halide is produced, for example:
  - **$Cl_2$ (g) + $H_2$ (g) → 2HCl (g)**
- These hydrogen halides react with ammonia gas to form **ammonium halides**
  - **$NH_3$ (g) + HCl (g) → $NH_4Cl$ (s)**
- Hydrogen halides will also react with water
- For example, hydrogen chloride also dissolves in water to form **hydrochloric acid**
  - **HCl (g) → $H^+$ (aq) + $Cl^-$ (aq)**

Your notes

# Fluorine & Astatine

**Fluorides and Astatides**

- If you are asked to predict reactions of fluorides and astatides that you are not familiar with you can use the information about Group 7 to help
- There are, however, exceptions to some of the reactions
  - For example, silver fluoride is soluble, so we can not used acidified silver nitrate to test for the presence of fluoride ions
    - $Ag^+$ (aq) + $F^-$ (aq) → AgF (aq)

### Worked Example

Write an equation to show the formation of an acid when hydrogen astatide is added to water.

**Answer:**

- Based on the reaction of hydrogen iodide with water:
  - HI (g) + $H_2O$ (g) → $H_3O^+$ (aq) + $I^-$ (aq)
  - So, hydrogen astatide must be:
    - **HAt (g) + $H_2O$ (g) → $H_3O^+$ (aq) + $At^-$ (aq)**